

CUSTOMER SHOPPING BEHAVIOUR ANALYSIS: AN END-TO-END DATA ANALYTICS PROJECT

CHAPTER 1: INTRODUCTION

1.1 Project Overview

In today's digital era, businesses generate massive amounts of customer transaction data through online and offline shopping platforms. Analyzing this data helps organizations understand customer preferences, purchasing behavior, spending patterns, and overall business performance. Data analytics enables companies to transform raw data into actionable insights that support strategic decision-making.

This project, **"Shopping Behaviour Analysis – End-to-End Data Analytics Project,"** focuses on analyzing customer shopping data using industry-standard tools such as Python, MySQL, and Power BI. The project follows a complete analytics pipeline, beginning with data cleaning and preprocessing, followed by SQL-based business analysis and interactive dashboard development.

The objective is to identify meaningful patterns in customer purchases, evaluate the impact of discounts and subscriptions, analyze product performance, and provide data-driven recommendations that can help improve business outcomes.

1.2 Problem Statement

Retail organizations often possess large volumes of customer transaction data but struggle to convert it into meaningful business insights. Without proper analysis, it becomes difficult to understand customer behavior, optimize marketing campaigns, improve customer retention, and maximize revenue generation.

This project addresses these challenges by analyzing customer shopping behavior data to answer important business questions related to customer spending habits, product performance, discount effectiveness, subscription trends, and revenue contribution across different customer segments.

1.3 Objectives

The primary objectives of this project are:

- To clean and preprocess raw customer shopping data using Python.
- To perform exploratory and business-focused data analysis using SQL.
- To identify customer purchasing patterns and spending behavior.

- To evaluate the impact of discounts on customer purchases.
- To analyze subscription-based customer engagement.
- To identify top-performing products and categories.
- To segment customers based on purchasing history.
- To create interactive Power BI dashboards for visual analysis.
- To generate actionable business insights and recommendations.

1.4 Scope of the Project

The scope of this project includes the analysis of customer shopping behavior using historical transaction data. The project covers:

- Data preprocessing and transformation.
- Customer demographic analysis.
- Revenue analysis.
- Product and category performance evaluation.
- Shipping preference analysis.
- Discount effectiveness assessment.
- Customer segmentation.
- Dashboard creation and reporting.

The project is limited to descriptive and diagnostic analytics and does not include predictive modeling or machine learning techniques.

1.5 Expected Outcomes

Upon completion of this project, the following outcomes are expected:

- Improved understanding of customer purchasing behavior.
- Identification of high-value customer segments.
- Recognition of top-performing products and categories.
- Insights into discount and subscription effectiveness.
- Interactive dashboards for business monitoring.
- Data-driven recommendations for improving customer engagement and revenue growth.

1.6 Project Workflow

The project follows a structured end-to-end analytics workflow:

1. Data Collection
2. Data Cleaning and Preprocessing using Python
3. Exploratory Data Analysis
4. Database Creation in MySQL
5. SQL-Based Business Analysis
6. Dashboard Development in Power BI
7. Insight Generation and Reporting
8. Documentation and GitHub Deployment

This workflow demonstrates practical implementation of data analytics concepts commonly used in industry environments.

Chapter 2: Tools & Technologies Used

2.1 Python

Used for data cleaning, preprocessing, and exploratory data analysis. Libraries such as Pandas and NumPy were utilized to handle missing values, transform data, and prepare the dataset for analysis.

2.2 MySQL

Used to store the cleaned dataset and perform business-focused analytical queries. SQL was used to extract insights related to customer behavior, revenue, discounts, subscriptions, and product performance.

2.3 Power BI

Used to build interactive dashboards and visualize key business metrics. It helped present trends, customer segments, product performance, and revenue insights in a user-friendly format.

2.4 Git & GitHub

Used for version control and project management. GitHub was used to maintain the project repository, track changes, and showcase the project portfolio professionally.

2.5 Jupyter Notebook

Used as the development environment for performing data cleaning, transformation, analysis, and documenting the workflow.

Technology Stack Summary

Category	Tools Used
Programming Language	Python
Data Processing	Pandas, NumPy
Database	MySQL
Data Visualization	Power BI
Development Environment	Jupyter Notebook
Version Control	Git, GitHub

Chapter 3: Dataset Description and Data Preparation

3.1 Dataset Overview

The project uses a customer shopping behavior dataset containing information about customer demographics, purchasing patterns, product preferences, discounts, subscriptions, shipping methods, and review ratings.

The dataset helps analyze customer behavior and identify factors influencing purchasing decisions.

3.2 Dataset Features

The dataset contains the following important attributes:

Attribute	Description
customer_id	Unique customer identifier
age	Customer age
gender	Male/Female
item_purchased	Product purchased
category	Product category
purchase_amount	Amount spent
location	Customer location
size	Product size
color	Product color
season	Season of purchase

review_rating	Product rating
subscription_status	Subscription status
shipping_type	Shipping method
discount_applied	Discount usage
previous_purchases	Number of previous purchases
payment_method	Payment method
frequency_of_purchases	Purchase frequency
age_group	Customer age segment

3.3 Data Cleaning and Preprocessing

Data preprocessing was performed using Python in Jupyter Notebook to ensure data quality and consistency.

The following steps were carried out:

- Imported dataset using Pandas.
- Checked dataset structure and data types.
- Identified and handled missing values.
- Removed duplicate records.
- Standardized categorical values.
- Verified numerical columns for consistency.
- Created derived fields for better analysis.
- Exported cleaned dataset for SQL and Power BI analysis.

3.4 Data Processing Workflow

The data preparation process followed the workflow below:

Raw Dataset → Data Cleaning → Data Validation → Feature Preparation → Cleaned Dataset → SQL Analysis → Power BI Dashboard

3.5 Output of Data Preparation

After preprocessing, a cleaned dataset was generated and used for:

- SQL-based business analysis
- Customer segmentation
- Revenue analysis

- Product performance analysis
- Power BI dashboard creation

Figure 3.1: Dataset Preview

Customer ID	Age	Gender	Item Purchased	Category	purchase amount	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method
1	55	Male	Blouse	Clothing	4505	Karnataka	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	PhonePe
2	19	Male	Sweater	Clothing	5440	Andhra Pradesh	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash
3	50	Male	Jeans	Clothing	6205	Telangana	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card
4	21	Male	Sandals	Footwear	7650	Karnataka	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	Google Pay
5	45	Male	Blouse	Clothing	4165	Tamil Nadu	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	Google Pay
6	46	Male	Sneakers	Footwear	1700	Kerala	M	White	Summer	2.9	Yes	Standard	Yes	Yes	14	PhonePe
7	63	Male	Shirt	Clothing	7225	Maharashtra	M	Gray	Fall	3.2	Yes	Free Shipping	Yes	Yes	49	Cash
8	27	Male	Shorts	Clothing	2890	Punjab	L	Charcoal	Winter	3.2	Yes	Free Shipping	Yes	Yes	19	Credit Card
9	26	Male	Coat	Outerwear	8245	Kerala	L	Silver	Summer	2.6	Yes	Express	Yes	Yes	8	PhonePe
10	57	Male	Handbag	Accessories	2635	Haryana	M	Pink	Spring	4.8	Yes	2-Day Ship	Yes	Yes	4	Cash
11	53	Male	Shoes	Footwear	2890	Rajasthan	L	Purple	Fall	4.1	Yes	Store Pick	Yes	Yes	26	Net Bank
12	30	Male	Shorts	Clothing	5780	Hawaii	S	Olive	Winter	4.9	Yes	Store Pick	Yes	Yes	10	Net Bank
13	61	Male	Coat	Outerwear	6120	Gujarat	M	Gold	Winter	4.5	Yes	Express	Yes	Yes	37	PhonePe
14	65	Male	Dress	Clothing	4335	New Hampshire	M	Violet	Spring	4.7	Yes	Express	Yes	Yes	31	Google Pay
15	64	Male	Coat	Outerwear	4505	New York	L	Teal	Winter	4.7	Yes	Free Shipping	Yes	Yes	34	Debit Card
16	64	Male	Skirt	Clothing	6885	Karnataka	M	Teal	Winter	2.8	Yes	Store Pick	Yes	Yes	8	Google Pay
17	25	Male	Sunglasses	Accessories	3060	Alabama	S	Gray	Spring	4.1	Yes	Next Day	Yes	Yes	44	Debit Card
18	53	Male	Dress	Clothing	3230	Mississippi	XL	Lavender	Winter	4.7	Yes	2-Day Ship	Yes	Yes	36	PhonePe
19	52	Male	Sweater	Clothing	4080	Maharashtra	S	Black	Summer	4.6	Yes	Free Shipping	Yes	Yes	17	Cash
20	66	Male	Pants	Clothing	7650	Karnataka	M	Green	Summer	3.3	Yes	Standard	Yes	Yes	46	Debit Card
21	21	Male	Pants	Clothing	4335	Punjab	M	Black	Winter	2.8	Yes	Express	Yes	Yes	50	Cash
22	31	Male	Pants	Clothing	5270	North Carolina	M	Charcoal	Winter	4.1	Yes	Store Pick	Yes	Yes	22	Debit Card
23	56	Male	Pants	Clothing	3145	Andhra Pradesh	M	Peach	Summer	3.2	Yes	Store Pick	Yes	Yes	32	Debit Card
24	31	Male	Pants	Clothing	7480	Odisha	XL	White	Winter	4.4	Yes	Express	Yes	Yes	40	Credit Card
25	18	Male	Jacket	Outerwear	1870	Tamil Nadu	M	Green	Fall	2.9	Yes	Store Pick	Yes	Yes	16	Debit Card
26	18	Male	Hoodie	Clothing	2125	Telangana	M	Silver	Summer	3.6	Yes	Express	Yes	Yes	14	Google Pay

Figure 3.2: Data Cleaning Process Screenshot

```
1 import pandas as pd
2 df = pd.read_csv("customer_shopping_behavior.csv")

1 df.head()

Customer ID Age Gender Item Purchased Category purchase amount Location Size Color Season Review Rating Subscription Status Shipping Type Discount Applied Promo Code Used Previous Purchases Payment Method
0 1 55 Male Blouse Clothing 4505 Karnataka L Gray Winter 3.1 Yes Express Yes Yes Yes 14 PhonePe
1 2 19 Male Sweater Clothing 5440 Andhra Pradesh L Maroon Winter 3.1 Yes Express Yes Yes Yes 2 Cash
2 3 50 Male Jeans Clothing 6205 Telangana S Maroon Spring 3.1 Yes Free Shipping Yes Yes Yes 23 Credit Card
3 4 21 Male Sandals Footwear 7650 Karnataka M Maroon Spring 3.5 Yes Next Day Air Yes Yes Yes 49 Google Pay
4 5 45 Male Blouse Clothing 4165 Tamil Nadu M Turquoise Spring 2.7 Yes Free Shipping Yes Yes Yes 31 Google Pay

1 df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
# Column Non-Null Count Dtype
-----
0 Customer ID 3900 non-null int64
1 Age 3900 non-null int64
2 Gender 3900 non-null object
3 Item Purchased 3900 non-null object
4 Category 3900 non-null object
5 purchase amount 3900 non-null float64
6 Location 3900 non-null object
7 Size 3900 non-null object
8 Color 3900 non-null object
9 Season 3900 non-null object
10 Review Rating 3900 non-null float64
11 Subscription Status 3900 non-null object
12 Shipping Type 3900 non-null object
13 Discount Applied 3900 non-null object
14 Promo Code Used 3900 non-null object
15 Previous Purchases 3900 non-null int64
16 Payment Method 3900 non-null object
```

Figure 3.3: Cleaned Dataset Output

customer_id	age	gender	item_purci	category	purchase_location	size	color	season	review_rat	subscrptic	shipping_t	discount	previous	payment	frequency	age_group	purchase_frequency	days
1	55	Male	Blouse	Clothing	4505 Karnataka	L	Gray	Winter	3.1	Yes	Express	Yes	14	PhonePe	Fortnightly	Middle-ag	14	
2	19	Male	Sweater	Clothing	5440 Andhra Pri	L	Maroon	Winter	3.1	Yes	Express	Yes	2	Cash	Fortnightly	Young-adu	14	
3	50	Male	Jeans	Clothing	6205 Telangana	S	Maroon	Spring	3.1	Yes	Free Shipp	Yes	23	Credit Car	Weekly	Middle-ag	7	
4	21	Male	Sandals	Footwear	7650 Karnataka	M	Maroon	Spring	3.5	Yes	Next Day /	Yes	49	Google Pa	Weekly	Young-adu	7	
5	45	Male	Blouse	Clothing	4165 Tamil Nad	M	Turquoise	Spring	2.7	Yes	Free Shipp	Yes	31	Google Pa	Annually	Middle-ag	365	
6	46	Male	Sneakers	Footwear	1700 Kerala	M	White	Summer	2.9	Yes	Standard	Yes	14	PhonePe	Weekly	Middle-ag	7	
7	63	Male	Shirt	Clothing	7225 Maharastr	M	Gray	Fall	3.2	Yes	Free Shipp	Yes	49	Cash	Quarterly	Seniors	90	
8	27	Male	Shorts	Clothing	2890 Punjab	L	Charcoal	Winter	3.2	Yes	Free Shipp	Yes	19	Credit Car	Weekly	Young-adu	7	
9	26	Male	Coat	Outerwear	8245 Kerala	L	Silver	Summer	2.6	Yes	Express	Yes	8	PhonePe	Annually	Young-adu	365	
10	57	Male	Handbag	Accessorie	2635 Haryana	M	Pink	Spring	4.8	Yes	2-Day Ship	Yes	4	Cash	Quarterly	Middle-ag	90	
11	53	Male	Shoes	Footwear	2890 Rajasthan	L	Purple	Fall	4.1	Yes	Store Pick	Yes	26	Net Bankir	Bi-Weekly	Middle-ag	14	
12	30	Male	Shorts	Clothing	5780 Hawaii	S	Olive	Winter	4.9	Yes	Store Pick	Yes	10	Net Bankir	Fortnightly	Young-adu	14	
13	61	Male	Coat	Outerwear	6120 Gujarat	M	Gold	Winter	4.5	Yes	Express	Yes	37	PhonePe	Fortnightly	Seniors	14	
14	65	Male	Dress	Clothing	4335 New Ham	M	Violet	Spring	4.7	Yes	Express	Yes	31	Google Pa	Weekly	Seniors	7	
15	64	Male	Coat	Outerwear	4505 New York	L	Teal	Winter	4.7	Yes	Free Shipp	Yes	34	Debit Card	Weekly	Seniors	7	
16	64	Male	Skirt	Clothing	6885 Karnataka	M	Teal	Winter	2.8	Yes	Store Pick	Yes	8	Google Pa	Monthly	Seniors	30	
17	25	Male	Sunglasses	Accessorie	3060 Alabama	S	Gray	Spring	4.1	Yes	Next Day /	Yes	44	Debit Card	Bi-Weekly	Young-adu	14	
18	53	Male	Dress	Clothing	3230 Mississippi	XL	Lavender	Winter	4.7	Yes	2-Day Ship	Yes	36	PhonePe	Quarterly	Middle-ag	90	
19	52	Male	Sweater	Clothing	4080 Maharastr	S	Black	Summer	4.6	Yes	Free Shipp	Yes	17	Cash	Weekly	Middle-ag	7	
20	66	Male	Pants	Clothing	7650 Karnataka	M	Green	Summer	3.3	Yes	Standard	Yes	46	Debit Card	Bi-Weekly	Seniors	14	
21	21	Male	Pants	Clothing	4335 Punjab	M	Black	Winter	2.8	Yes	Express	Yes	50	Cash	Every 3 M	Young-adu	90	
22	31	Male	Pants	Clothing	5270 North Can	M	Charcoal	Winter	4.1	Yes	Store Pick	Yes	22	Debit Card	Quarterly	Young-adu	90	
23	56	Male	Pants	Clothing	3145 Andhra Pri	M	Peach	Summer	3.2	Yes	Store Pick	Yes	32	Debit Card	Annually	Middle-ag	365	
24	31	Male	Pants	Clothing	7480 Odisha	XL	White	Winter	4.4	Yes	Express	Yes	40	Credit Car	Weekly	Young-adu	7	
25	18	Male	Jacket	Outerwear	1870 Tamil Nad	M	Green	Fall	2.9	Yes	Store Pick	Yes	16	Debit Card	Weekly	Young-adu	7	

Chapter 4: SQL-Based Business Analysis

4.1 Introduction

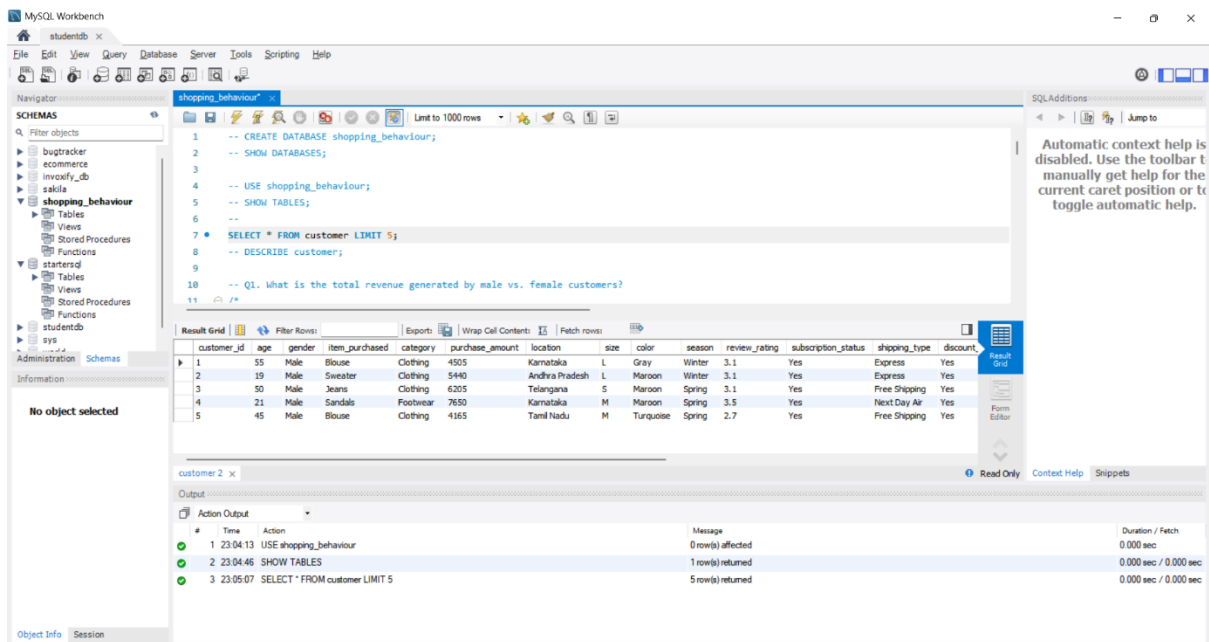
After completing data cleaning and preprocessing, the cleaned dataset was imported into MySQL Workbench for analytical processing. SQL queries were used to answer key business questions related to customer behavior, revenue generation, product performance, subscriptions, discounts, and customer segmentation.

The objective of this phase was to transform raw transactional data into meaningful business insights that could support strategic decision-making.

4.2 Database Setup

A database named **shopping_behaviour** was created in MySQL Workbench. The cleaned dataset was imported into a table named **customer** using the MySQL Import Wizard.

The imported table contained all cleaned and transformed attributes required for business analysis.



4.3 Revenue Analysis by Gender

Business Question

Which gender contributes more to overall revenue?

Purpose

To identify revenue contribution across customer demographics.

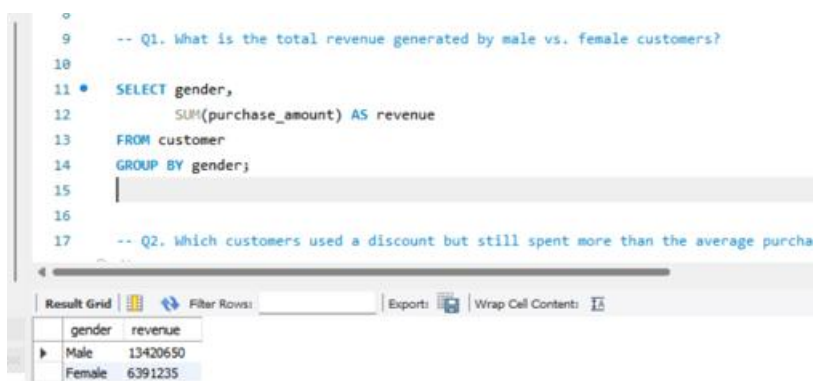
Observation

The analysis showed that male customers generated higher overall revenue compared to female customers.

Business Insight

Understanding demographic spending patterns helps businesses design targeted marketing campaigns and promotional strategies.

Figure 4.1: Revenue by Gender Query Output



4.4 High-Value Customers Using Discounts

Business Question

Which customers used discounts but still spent above the average purchase amount?

Purpose

To identify valuable customers who remain profitable despite receiving discounts.

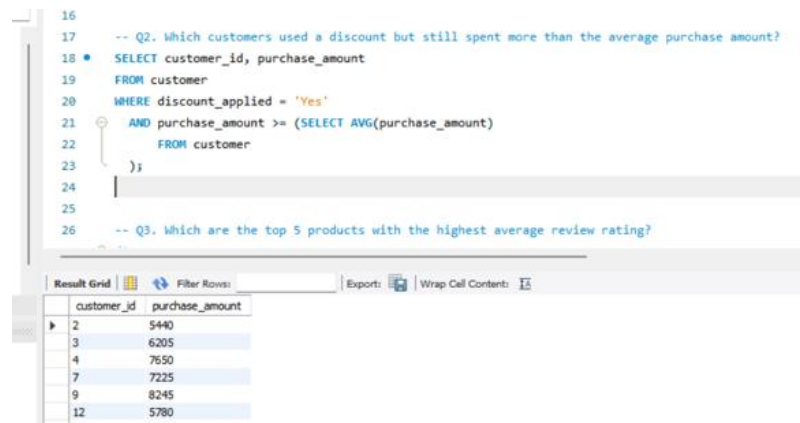
Observation

Several customers continued to spend significantly above average even when discounts were applied.

Business Insight

These customers can be targeted through loyalty programs and personalized offers.

Figure 4.2: High-Value Discount Customers Output



```
16
17 -- Q2. Which customers used a discount but still spent more than the average purchase amount?
18 • SELECT customer_id, purchase_amount
19 FROM customer
20 WHERE discount_applied = 'Yes'
21 AND purchase_amount >= (SELECT AVG(purchase_amount)
22 FROM customer
23 );
24
25
26 -- Q3. Which are the top 5 products with the highest average review rating?
```

customer_id	purchase_amount
2	9440
3	6205
4	7650
7	7225
9	8245
12	5780

4.5 Top Rated Products

Business Question

Which products have the highest average review ratings?

Purpose

To identify products that receive the highest customer satisfaction.

Observation

Products such as Gloves, Sandals, Boots, Hats, and Handbags achieved the highest average ratings.

Business Insight

Highly rated products should be prioritized in promotions and inventory planning.

Figure 4.3: Top Rated Products Output

40
27 -- Q3. Which are the top 5 products with the highest average review rating?
28
29 • SELECT item_purchased, ROUND(AVG(review_rating), 2) AS avg_product_rating
30 FROM customer
31 GROUP BY item_purchased
32 ORDER BY avg_product_rating DESC
33 LIMIT 5;
34
35

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

item_purchased	avg_product_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Handbag	3.78

4.6 Shipping Preference Analysis

Business Question

How does purchase behavior differ across shipping methods?

Purpose

To understand customer preferences for shipping services.

Observation

Customers using Express shipping generally recorded slightly higher purchase values.

Business Insight

Premium shipping services can be marketed to high-value customers.

Figure 4.4: Shipping Analysis Output

35
36 -- Q4. Compare the average Purchase Amounts between Standard and Express Shipping.
37
38 • SELECT shipping_type, ROUND(AVG(purchase_amount), 2) AS avg_purchase_amount
39 FROM customer
40 WHERE shipping_type IN ('Standard', 'Express')
41 GROUP BY shipping_type;
42
43
44 -- Q5. Do subscribed customers spend more? Compare average spend and total revenue between

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

shipping_type	avg_purchase_amount
Express	5140.39
Standard	4969.12

4.7 Subscription-Based Customer Analysis

Business Question

Do subscribed customers spend more than non-subscribed customers?

Purpose

To evaluate the effectiveness of the subscription program.

Observation

Subscribers demonstrated higher average spending and stronger revenue contribution.

Business Insight

Subscription programs can significantly improve customer lifetime value.

Figure 4.5: Subscription Analysis Output

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

4.8 Product Discount Analysis

Business Question

Which products receive the highest percentage of discount-based purchases?

Purpose

To understand product dependency on discounts.

Observation

Certain products showed a high reliance on discount-driven purchases.

Business Insight

Businesses should carefully monitor discount strategies to protect profit margins.

Figure 4.6: Discount Analysis Output

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

4.9 Customer Segmentation

Business Question

How can customers be segmented based on purchase history?

Purpose

To categorize customers according to loyalty and purchasing frequency.

Segments Created

Segment Definition

New 1 Previous Purchase

Returning 2–10 Previous Purchases

Loyal More than 10 Previous Purchases

Business Insight

Different customer segments require different engagement strategies.

Figure 4.7: Customer Segmentation Output

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

4.10 Category-Wise Product Performance

Business Question

What are the top-performing products within each category?

Purpose

To identify category-level best sellers.

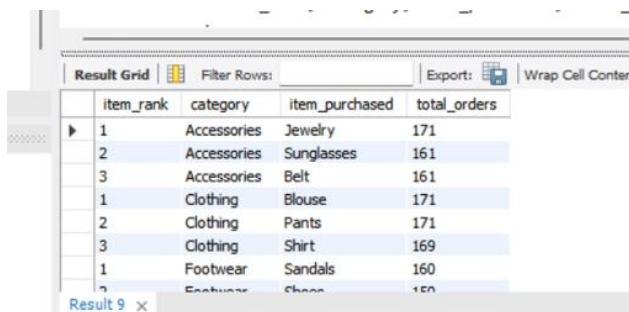
Observation

Several products consistently outperformed others within their respective categories.

Business Insight

These products should receive greater inventory allocation and promotional focus.

Figure 4.8: Category Performance Output



The screenshot shows a data table with columns: item_rank, category, item_purchased, and total_orders. The table lists the top-performing products within each category. The categories are Accessories, Clothing, and Footwear. The top-performing products are Jewelry, Sunglasses, Belt, Blouse, Pants, Shirt, Sandals, and Shoes.

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150

4.11 Repeat Buyer and Subscription Relationship

Business Question

Are repeat buyers more likely to subscribe?

Purpose

To understand the relationship between purchase frequency and subscriptions.

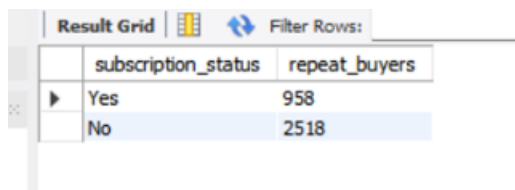
Observation

A significant number of repeat buyers were still non-subscribers.

Business Insight

This group represents an excellent opportunity for subscription conversion campaigns.

Figure 4.9: Repeat Buyer Analysis Output



The screenshot shows a data table with two columns: 'subscription_status' and 'repeat_buyers'. The 'subscription_status' column has two rows: 'Yes' and 'No'. The 'repeat_buyers' column has two rows: '958' and '2518'. The table is displayed in a software interface with a 'Result Grid' header and a 'Filter Rows' button.

subscription_status	repeat_buyers
Yes	958
No	2518

4.12 Revenue Contribution by Age Group

Business Question

Which age group contributes the highest revenue?

Purpose

To identify the most valuable customer age segment.

Observation

Young Adults generated the highest revenue contribution, followed by Middle-Aged customers.

Business Insight

Marketing campaigns can be optimized by focusing on high-value demographic groups.

Figure 4.10: Revenue by Age Group Output

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

4.13 Summary of SQL Findings

The SQL analysis revealed several important insights:

- Male customers generated higher revenue.
- Subscribers contributed greater business value.
- Young Adults formed the strongest revenue-generating segment.
- High-rated products demonstrated strong customer satisfaction.
- Repeat buyers represent an opportunity for subscription growth.
- Discount strategies significantly influence purchasing behavior.
- Customer segmentation enables targeted marketing and retention initiatives.

Perfect. For a professional portfolio report, **one full dashboard screenshot is actually better** than showing 8–10 individual visual screenshots. It looks cleaner and more professional.

Chapter 5: Power BI Dashboard Development

5.1 Introduction

After completing data cleaning in Jupyter Notebook and business analysis in MySQL Workbench, the cleaned dataset was imported into Microsoft Power BI for dashboard creation and interactive visualization.

The objective of this stage was to transform analytical results into a visually intuitive dashboard that allows users to monitor customer behavior, revenue trends, subscription patterns, and product performance in a single view.

5.2 Data Integration

The cleaned dataset, **customer_behavior_cleaned_aftershoppinganalysis.csv**, was imported into Power BI Desktop using the Text/CSV connector.

Before creating visualizations, the following validation steps were performed:

- Verified column names and data types.
- Checked numerical fields for consistency.
- Confirmed categorical attributes were properly classified.
- Validated newly engineered columns such as Age Group and Purchase Frequency Days.

The dataset was then loaded into the Power BI data model for reporting and visualization.

5.3 Dashboard Design Objectives

The dashboard was designed to provide answers to the following business questions:

- How many customers are present in the dataset?

- What is the average purchase amount?
- How are customers distributed across subscription categories?
- Which product categories generate the highest revenue?
- Which age groups contribute most to sales?
- What purchasing patterns can be observed across customer segments?

The design focuses on simplicity, readability, and business relevance.

5.4 Key Performance Indicators (KPIs)

To provide a quick overview of business performance, several KPI cards were included at the top of the dashboard.

Total Customers

Displays the total number of customers available in the dataset.

Business Value:

Provides an overall understanding of the customer base size.

Average Purchase Amount

Displays the average amount spent per transaction.

Business Value:

Helps evaluate customer spending behavior and purchasing power.

Average Review Rating

Displays the average customer review score.

Business Value:

Measures overall customer satisfaction and product acceptance.

5.5 Dashboard Visualizations

Customer Subscription Status

A pie chart was used to visualize the distribution of subscribed and non-subscribed customers.

Observation

The majority of customers were non-subscribers, while a smaller segment had active subscriptions.

Business Insight

The large non-subscriber segment presents an opportunity for subscription conversion campaigns.

Revenue by Category

A bar chart was created to compare revenue generated by different product categories.

Observation

Clothing generated the highest revenue, followed by Accessories and Footwear.

Business Insight

High-performing categories should receive greater marketing and inventory investment.

Sales by Category

This visual shows the number of purchases within each product category.

Observation

Customer demand was concentrated in a few major categories.

Business Insight

Understanding category popularity helps optimize product assortment and stock allocation.

Revenue by Age Group

A bar chart was used to analyze revenue contribution across age segments.

Observation

Young Adults generated the highest revenue contribution.

Business Insight

Marketing campaigns can be optimized by focusing on the most valuable demographic groups.

Sales by Age Group

This visual compares transaction volume across age groups.

Observation

Young Adults recorded the highest purchase frequency.

Business Insight

This segment represents a strong opportunity for retention and loyalty programs.

5.6 Interactive Dashboard Features

To improve user experience and analytical flexibility, interactive slicers were implemented.

Slicers Used

- Gender
- Category
- Subscription Status
- Shipping Type

These filters allow users to dynamically explore data and generate customized insights.

5.7 Dashboard Screenshot

Figure 5.1: Customer Shopping Behaviour Power BI Dashboard



The dashboard provides a centralized view of customer behavior, revenue trends, subscription distribution, category performance, and demographic insights.

5.8 Benefits of the Dashboard

The Power BI dashboard enables:

- Faster business decision-making.
- Real-time monitoring of customer metrics.
- Identification of high-performing product categories.
- Better understanding of customer demographics.
- Improved reporting and stakeholder communication.

5.9 Summary

The Power BI dashboard successfully transformed raw analytical results into meaningful business intelligence. By combining KPI cards, charts, and interactive filters, the dashboard provides a comprehensive view of customer shopping behavior and supports data-driven decision-making.

Chapter 6: Key Findings and Business Recommendations

6.1 Key Findings

The analysis of customer shopping behavior revealed several valuable insights related to customer demographics, purchasing patterns, product performance, and subscription behavior.

Customer Demographics

- Male customers contributed higher overall revenue compared to female customers.
- Young Adult customers generated the highest revenue and sales volume among all age groups.
- Customer purchasing behavior varied significantly across demographic segments.

Customer Loyalty and Subscription Analysis

- A large portion of customers were identified as non-subscribers.
- Subscribers demonstrated stronger spending behavior and higher customer value.
- Many repeat customers had not yet subscribed, indicating potential opportunities for customer conversion programs.

Product Performance

- Certain products consistently achieved higher customer ratings.
- Product categories such as Clothing and Accessories generated the highest revenue.
- High-performing products can be prioritized for promotional campaigns and inventory planning.

Discount Analysis

- Discounts influenced purchasing behavior across multiple product categories.
- Several customers continued to spend above average despite receiving discounts.
- Strategic discounting can increase sales without significantly impacting revenue generation.

Shipping Preferences

- Customers selecting premium shipping options showed slightly higher purchase values.
- Shipping preferences may serve as indicators of customer purchasing power and buying intent.

Dashboard Insights

The Power BI dashboard provided a consolidated view of customer behavior, revenue trends, product performance, and subscription status, enabling quick and effective business analysis.

6.2 Business Recommendations

Based on the analysis performed throughout this project, the following recommendations are proposed:

Strengthen Customer Loyalty Programs

Develop targeted loyalty initiatives for returning and high-value customers to improve retention and increase lifetime value.

Increase Subscription Adoption

Design promotional campaigns and exclusive benefits to encourage non-subscribers and repeat buyers to join subscription programs.

Focus on High-Performing Categories

Allocate additional marketing resources and inventory investment to categories generating the highest revenue and customer engagement.

Optimize Discount Strategies

Apply discounts selectively to maximize customer acquisition while protecting profit margins.

Enhance Customer Segmentation

Use customer segmentation to personalize marketing campaigns and improve customer experiences.

Leverage Dashboard Reporting

Utilize the Power BI dashboard as a decision-support tool for monitoring business performance and identifying emerging trends.

Chapter 7: Conclusion

7.1 Project Summary

This project successfully implemented a complete end-to-end data analytics workflow using Python, MySQL Workbench, and Power BI. Beginning with raw customer shopping data, the

project applied data cleaning, preprocessing, feature engineering, business-oriented SQL analysis, and interactive dashboard development to generate meaningful insights.

The analysis provided valuable information regarding customer demographics, purchasing behavior, subscription patterns, discount effectiveness, product performance, and revenue generation. These insights can support data-driven business decisions and contribute to improved customer engagement and operational efficiency.

7.2 Learning Outcomes

Through this project, practical experience was gained in:

- Data Cleaning and Preprocessing using Python
- Exploratory Data Analysis
- SQL Query Development and Business Analysis
- Data Visualization using Power BI
- Dashboard Design and Reporting
- Version Control and Project Documentation using GitHub

7.3 Future Enhancements

Future improvements to this project may include:

- Predictive analytics using Machine Learning models.
- Customer churn prediction.
- Product recommendation systems.
- Real-time dashboard integration.
- Advanced customer segmentation techniques.

7.4 Final Remarks

This project demonstrates how data analytics tools can transform raw transactional data into actionable business insights. By combining Python, SQL, and Power BI, the project delivers a practical and industry-relevant solution for understanding customer shopping behavior and supporting strategic decision-making.